INCH-POUND

MIL-DTL-81581/2B(AS)
24 July 2006
SUPERSEDING
MIL-H-81581/2A(AS)
01 June 1982

DETAIL SPECIFICATION SHEET

HOSE ASSEMBLIES, BREATHING OXYGEN, HIGH PRESSURE, WITH CONNECTIONS FOR AN INTEGRATED COMMUNICATION SYSTEM

Reactivated after 24 July 2006 and may be used for new and existing designs and acquisitions.

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-81581.

1. SCOPE

1.1 Scope. This specification sheet covers the detail requirements for high pressure, breathing oxygen hose assemblies with connections for integrated communication systems.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specification forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Code 491000B120-3, Highway 547, Lakehurst, NJ 08733-5100 or emailed to thomas.omara@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-81581	-	Hose Assemblies, Breathing Oxygen and Air; General Specification for

(Copies of the above specification are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following drawing forms a part of this document to the extent specified herein. Unless otherwise specified, the issue of this document is that cited in the solicitation or contract.

NAVAL AIR SYSTEMS COMMAND (NAVAIR) DRAWINGS

278AS100	-	Hose Assembly Index, Breathing Oxygen, High Pressure with Connections for an Integrated Communication System.

(Copies of the above drawing are available from the Naval Air Technical Data and Engineering Service Command (NATEC), P.O. Box 357031, NASNI, Bldg. 90, San Diego, CA 92135-7031 or at https://webjedmics.natec.navy.mil.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-DTL-81581 and as specified herein.

3.2 Qualification. The hose furnished under this specification shall be a product that is authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.1.1). Qualification shall be in accordance with groups as listed on NAVAIR Index Drawing 278AS100. Qualification approval of any individual hose listed within a group shall automatically qualify a manufacturer for all other items listed within the group provided the conditions set forth in MIL-DTL-81581, 6.3, Qualification, are complied with.

3.3 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.2.

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3.4 Talc. Talc shall not be used in the manufacturing process of the hoses furnished under this specification sheet.

3.5 Design and construction. The hose assembly shall be in accordance with the general specification and with the applicable drawing, NAVAIR 278AS100.

3.6 Performance. Unless otherwise specified herein, the performance of the hose assembly shall be as specified in the general specification, consistent with the applicable performance requirements for the applicable tests and inspections specified herein.

3.6.1 Polarity (color code) and electrical continuity. When tested as specified in the general specification, the resistance measured between all points designed to be electrically connected shall be as specified in the general specification with an allowable resistance of 0.075 ohms per foot length of the electrically wired portion of the hose.

3.6.2 Elongation.

3.6.2.1 Procedure I. When tested as specified in the general specification, the increase in length of the hose shall be not greater than 12 percent of the original hose length.

3.6.2.2 Procedure II. When tested as specified in the general specification, the permanent set of the hose shall be not greater than 2 percent of the original hose length when measured 2 minutes after removal of load.

3.6.3 Strength.

3.6.3.1 Tensile load. When tested as specified in the general specification, there shall be no indication of separation of the end fittings or molded ends from the hose assembly.

3.6.3.2. Static load. When tested as specified in the general specification, the hose shall not show a deflection of greater than 25 percent of original value on the outside diameter. With the load removed, the outside diameter of the hose shall be within 5 percent of original value.

3.6.4 Restraint cord elongation and tensile strength. When the applicable control drawing specifies a restraint cord, the restraint cord shall have a 30 percent maximum elongation and a minimum tensile strength of 70 pounds when tested as specified in the general specification.

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with the general specification and as specified herein.

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4.1.1 Qualification inspection. The qualification inspection of the hose assembly shall consist of the following examinations and tests from the general specification:

Visual examination
Polarity (color code) and electrical continuity, procedure I
Insulation resistance, procedure I
Elongation, procedure II
Odor, procedure I
Cleanliness, procedures I and II
Leakage, procedure III
Flexibility, procedure IV (under environmental conditions, procedure II)
Minimum burst pressure, procedure III
Tensile load, procedure II and procedure III (if applicable)
Static load, procedure I
Restraint cord elongation and tensile strength
Low temperature, procedure I
High temperature, procedure I
Flexibility endurance, procedure II
Ozone-resistance, procedure I
Abrasion, procedure I

4.1.2 First article inspection. The first article inspection of the hose assembly shall consist of the applicable examinations and tests specified in the general specification with the applicable procedure as specified herein.

4.1.2.1 First article samples. The first article samples shall consist of four hose assemblies of the part number contracted for.

4.1.2 Conformance inspection. Conformance inspection shall consist of the following tests from the general specification. The sample size and acceptance criteria shall be as specified in the general specification.

Visual examination
Polarity (color code) and electrical continuity, procedure I
Insulation resistance, procedure I
Elongation, procedure II
Odor, procedure I
Cleanliness, procedure I
Leakage, procedure III
Flexibility, procedure IV

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5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the general specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The hose assemblies covered by this detail specification sheet are intended for use in breathing oxygen, and contain integrated communication services for aircrewmen in the operation of the aircraft.

6.2 Acquisition requirements. Acquisition requirements should be in accordance with the general specification.

6.3 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
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